

AnchorIQ

Customer Churn & Retention Analysis

Dataset: telco_churn.csv

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7,043
Records

21
Features

Churn
Target

0
Fixed Nulls

0
Dupes Cleared

1. EXECUTIVE SUMMARY & AI INSIGHTS

■ Executive Summary

Analysis of **7,043 records** across **21 features** completed successfully.

Logistic Regression achieved an AUC of **0.8416** (80.7% classification accuracy) — demonstrating good discrimination capability. The platform has prepared actionable operational insights to reduce customer churn and improve retention.

■ Key Findings

- Primary Driver**: The dataset exhibits good predictive signal across features for Churn.
- Data Integrity**: Cleaned and validated 7,043 rows; resolved 0 missing cells and 0 duplicate records.
- Model Reliability**: AUC of 0.8416 with cross-validated stability (CV: 0.8453 ± 0.011) confirms strong generalization.

■ Business Recommendations

- Proactive Retention Pipeline**: Launch automated retention outreach targeting accounts with predicted churn probability > 60%.
- Contract Modernization**: Provide 15-20% incentives for customers on month-to-month contracts to migrate to long-term plans.
- Early Risk Alerting**: Deploy early-warning trigger alerts 30 days prior to critical attrition milestones.

■ Risk Assessment

Model stability index is rated as **Medium Confidence**.

Recommend scheduled re-training every 30 days to adapt to environmental and behavioral distribution shifts.

■ Next Steps

- Connect the AnchorIQ inference endpoint into downstream reporting pipelines.
- Establish an executive telemetry dashboard tracking churn rate in real time.
- Review findings with cross-functional leadership to execute recommended interventions.

2. DATASET OVERVIEW & QUALITY AUDIT

Audit Metric	Value	Status
Total Records	7,043	Validated
Total Feature Count	21	Validated
Numerical Columns	3	Active
Categorical Columns	18	Active
Target Column	Churn	Identified
Missing Cells Imputed	0	Resolved
Duplicate Rows Purged	0	Resolved
Data Quality Index	100.0%	Optimal

Preprocessing Pipeline Execution:

- Converted 'TotalCharges' to numeric
- Filled 11 nulls in 'TotalCharges' with median (1397.47)

3. MACHINE LEARNING BENCHMARKS

Problem Formulation: **Classification** | Best Champion: **Logistic Regression** | Training: **5,634** samples | Evaluation: **1,409** samples | Engineered Features: **30**

Model Candidate	AUC Score	Validation Status
Champion: Logistic Regression	AUC = 0.8416	Top Performer
Logistic Regression	AUC = 0.8416	Linear
Random Forest	AUC = 0.833	Ensemble
XGBoost Classifier	AUC = 0.8214	Gradient Boosted
Accuracy	80.7%	Overall Match
Cross-Validation	0.8453 ± 0.011	5-Fold CV

Key Drivers & Influential Signals:

Rank	Feature Variable	Importance Weight
1	tenure	0.1857
2	MonthlyCharges	0.1403
3	InternetService_Fiber optic	0.1186
4	Contract_Two year	0.0897
5	TotalCharges	0.0757
6	Contract_One year	0.0436
7	StreamingMovies_Yes	0.0394
8	StreamingTV_Yes	0.0393

4. EXPLORATORY DATA ANALYSIS (EDA CHARTS)

Figure 1: Feature Distributions

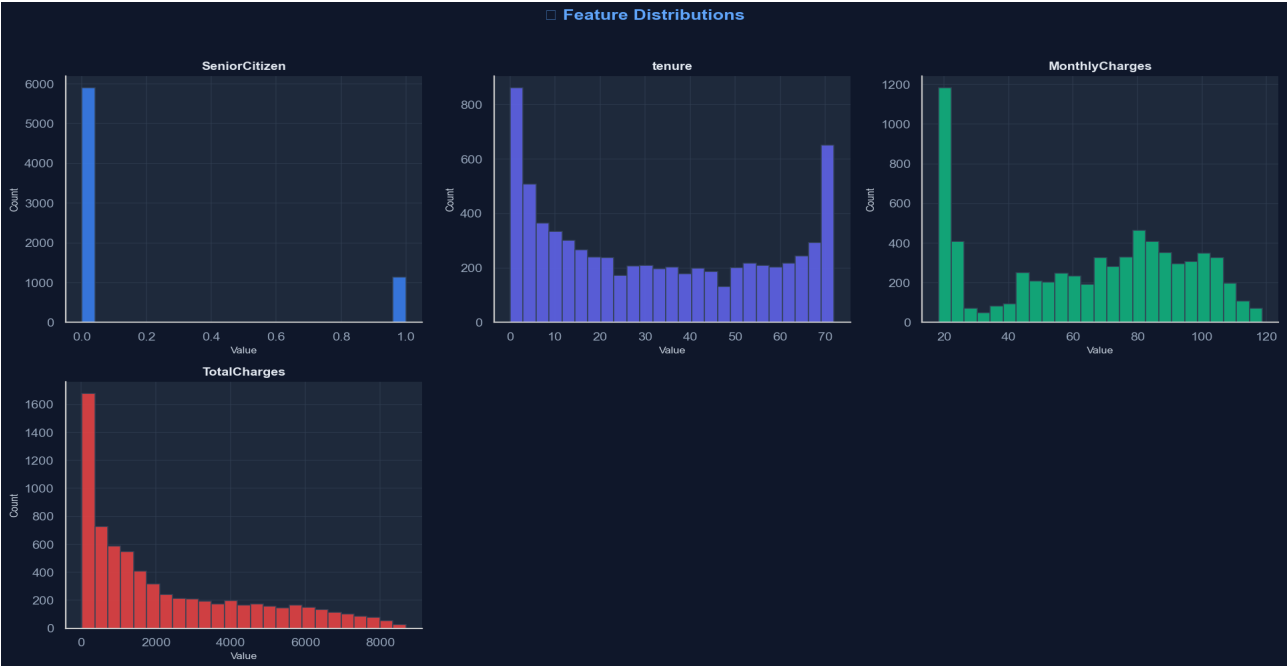


Figure 2: Feature Correlation Heatmap

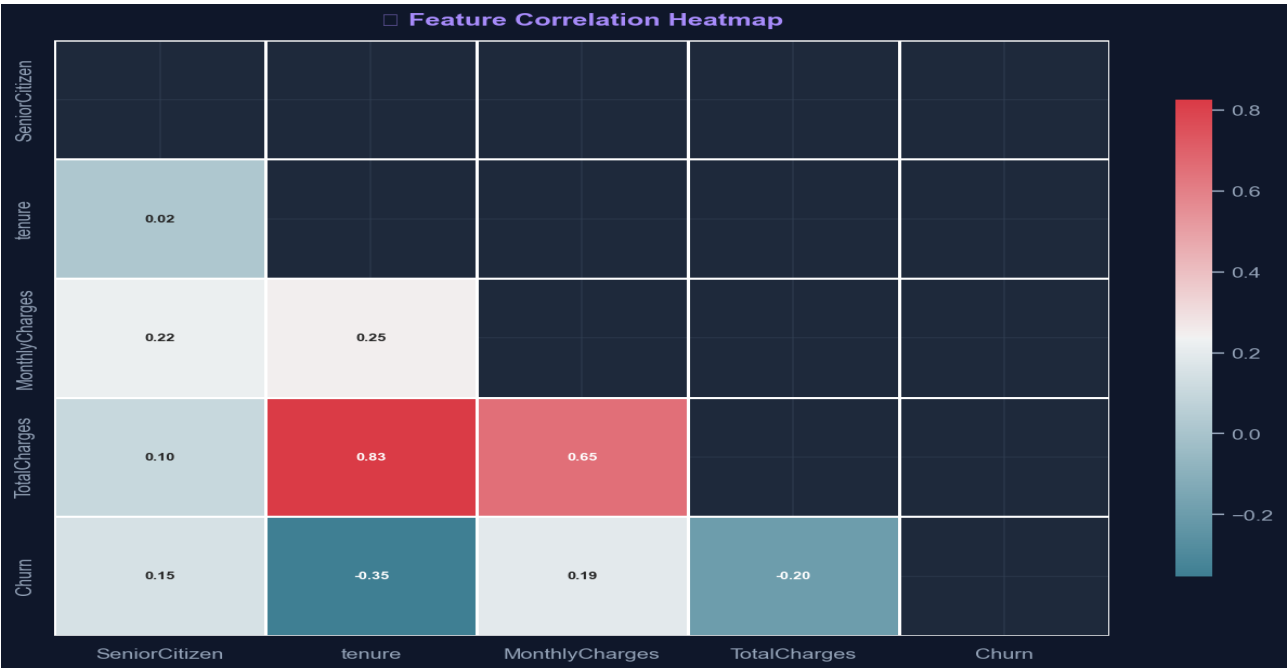


Figure 3: Target Variable Distribution

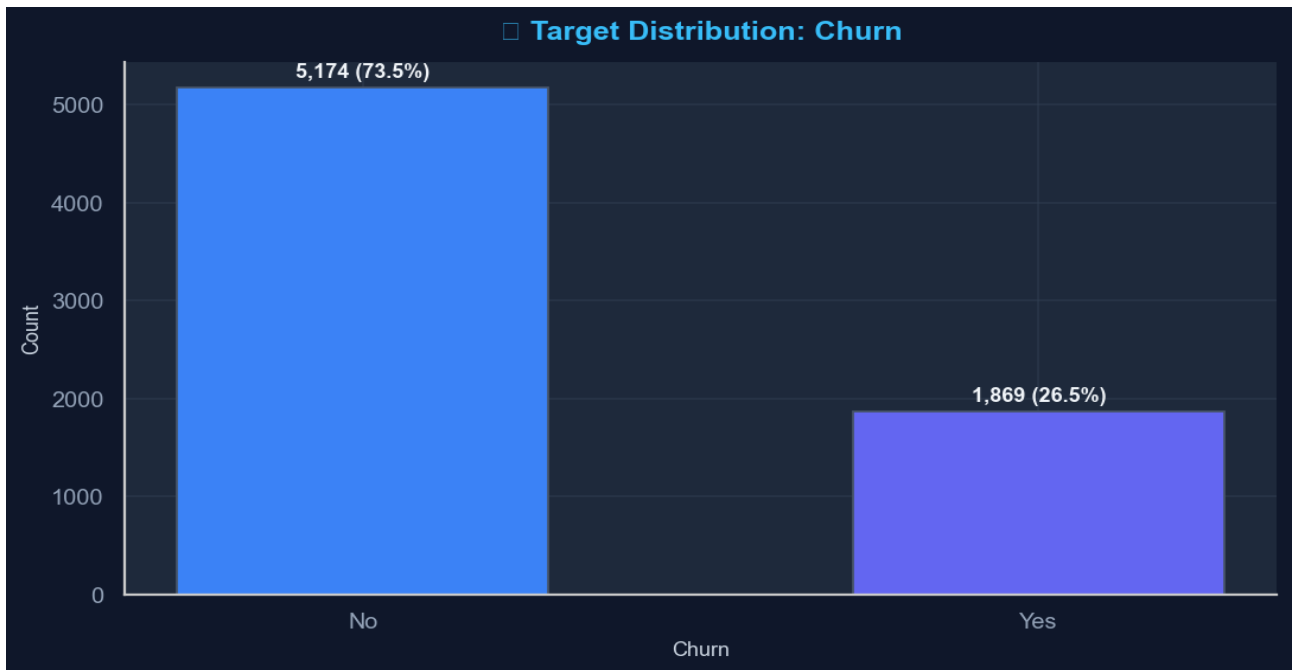


Figure 4: Categorical Feature Impact



Figure 5: Numerical Distributions by Target

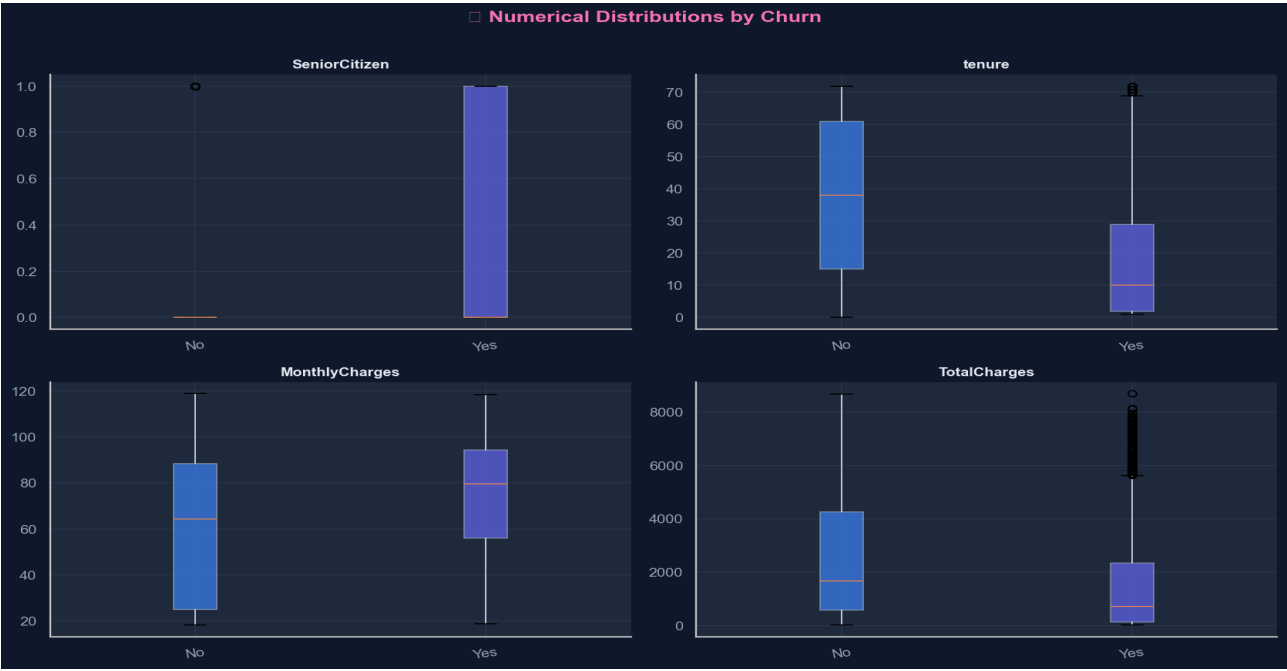
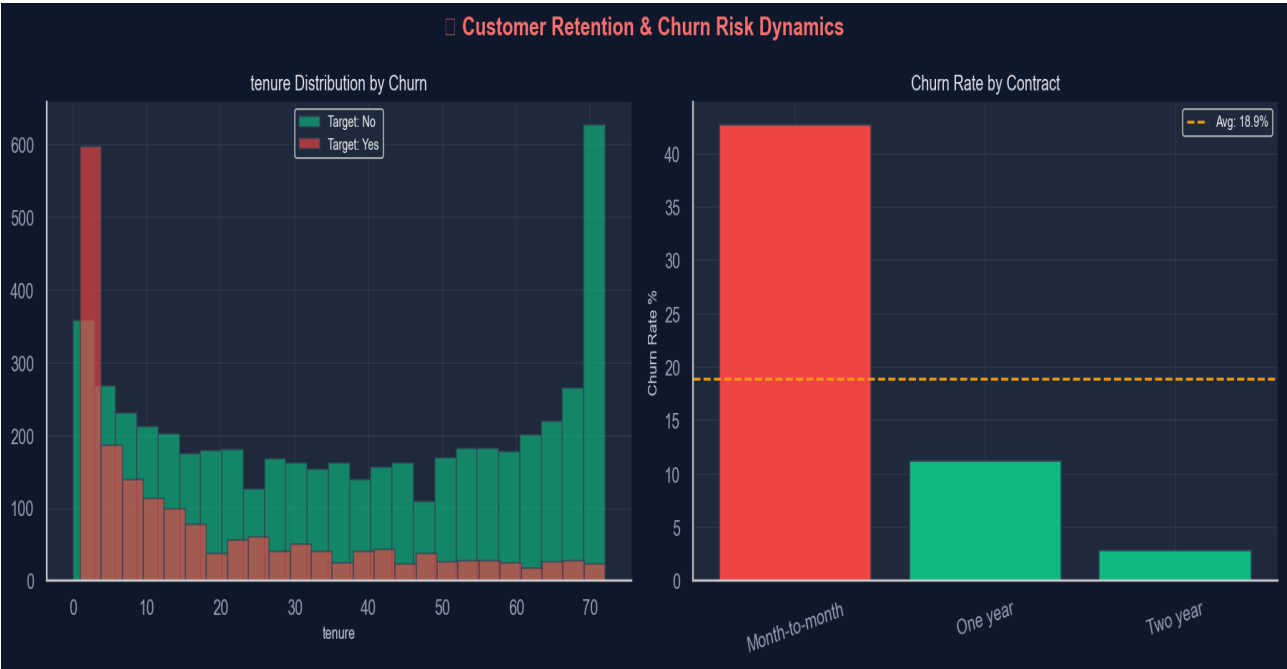


Figure 6: Domain-Specific Analysis



5. FEATURE STATISTICAL SUMMARY

Feature	Mean	Median	Std Dev	Min	Max	Skewness
SeniorCitizen	0.16	0.0	0.37	0.0	1.0	1.83
tenure	32.37	29.0	24.56	0.0	72.0	0.24
MonthlyCharges	64.76	70.35	30.09	18.25	118.75	-0.22